

Friedrich Collins

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EDUCATION

University of Cambridge | *MSc Computer Science (Specialization: Cybersecurity)* 2019 - 2020

- Completed dissertation on Zero-Trust Architecture implementation in distributed systems, achieving distinction grade with 78% average
- Led weekly security research seminars for 30+ postgraduate students covering threat modeling and cryptographic protocols

Georgia Institute of Technology | *BS Computer Science (Minor in Information Security)* 2015 - 2019

- Graduated summa cum laude with 3.88 GPA; recipient of Cybersecurity Excellence Award
- Co-founded Georgia Tech Ethical Hacking Club, growing membership to 150+ students within 18 months

EXPERIENCE

Cloudflare | *Security Engineer II* Jun 2022 - Present

- Architected and deployed a new API rate-limiting module that reduced DDoS attack surface by 34%, protecting 2M+ customer domains
- Designed automated vulnerability scanning pipeline integrating SAST/DAST tools, reducing security review time from 4 hours to 18 minutes per release
- Responded to 47 critical security incidents, maintaining 99.2% MTTR (mean time to response) and preventing \$2.3M in potential data exposure

Stripe | *Senior Security Analyst* Mar 2021 - May 2022

- Led penetration testing program across 8 microservices, identifying and remediating 156 vulnerabilities before production, representing 67% reduction in security debt
- Implemented OAuth 2.0 and JWT token validation framework processing 450K+ authentication requests daily with zero compromise incidents
- Mentored 3 junior security analysts, resulting in 2 promotions and 40% improvement in team incident resolution efficiency

Palantir Technologies | *Security Operations Analyst* Aug 2020 - Feb 2021

- Developed Python-based log aggregation tool correlating events from 12 security tools, reducing SIEM query time from 8 hours to 12 minutes
- Conducted security compliance audits achieving SOC 2 Type II certification with 98.4% control effectiveness score
- Configured and maintained intrusion detection system detecting 89 suspicious network patterns monthly with 92% true positive rate

PROJECTS

CryptVault - Zero-Trust Secret Management | *Go, gRPC, HashiCorp Vault, Docker, Kubernetes* Jan 2023 - Jun 2023

- Built distributed secret management platform using zero-trust principles, supporting 50K secrets with 99.99% availability and AES-256 encryption
- Reduced secret rotation time from 45 minutes to 2 minutes, eliminating manual operational overhead for 200+ developers across 15 teams

ThreatMapper - Continuous Vulnerability Intelligence | *Python, FastAPI, PostgreSQL, Shodan API, Docker* Sep 2022 - Dec 2022

- Engineered real-time threat intelligence aggregator scanning 45+ public vulnerability databases, generating 78% reduction in vulnerability discovery time
- Deployed alert system notifying 12 teams of critical CVE matches within 90 seconds, preventing 3 security incidents

SecAudit - Automated Compliance Framework | *Terraform, Python, AWS IAM, GitHub Actions, CloudFormation* Apr 2022 - Aug 2022

- Created Infrastructure-as-Code compliance checking tool validating 156 security policies across 40+ AWS accounts monthly
- Automated compliance reporting reducing manual audit work by 88%, saving 120 hours of manual labor quarterly

EncryptedLog - SIEM Log Encryption Suite | *Java, Spring Boot, ELK Stack, GPG, Kubernetes* Nov 2021 - Mar 2022

- Developed end-to-end encrypted logging system processing 2.1M log events daily while maintaining 99.97% uptime
- Reduced log analysis performance overhead by 12% through optimized encryption algorithms, enabling faster threat detection

EXTRACURRICULAR

OWASP (Open Web Application Security Project) | *Core Team Member & Chapter Lead* 2021 - Present

- Organized 24 security workshops reaching 680+ professionals, covering API security, threat modeling, and secure coding practices
- Maintained OWASP Top 10 educational resources repository with 4.2K GitHub stars and 89K monthly downloads

DEF CON Security Conference | *Speaker & Capture The Flag (CTF) Judge* 2022 - 2023

- Delivered keynote on 'Zero-Trust Architecture in Cloud Environments' to 1,200+ security professionals at DEF CON 31
- Designed and judged CTF challenges for 420 participants, with 94% participant satisfaction rating

TECHNICAL SKILLS

Languages – Python, Go, Java, Bash, SQL, JavaScript

Technologies – Kubernetes, Docker, AWS (IAM, CloudTrail, Security Hub), Azure, GCP, Terraform, SIEM (Splunk, ELK), Vault, OAuth 2.0, TLS/SSL, GPG

Tools – Burp Suite, Metasploit, Wireshark, Nessus, Snort, OWASP ZAP, Git, Jenkins, GitHub Actions, Terraform, Shodan